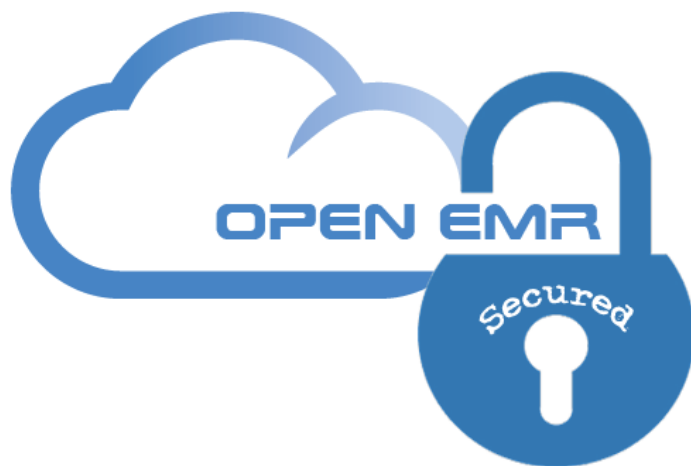




Hardening OpenEMR in Cloud (AWS)



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1 Introduction

This Technical Guide is intended to provide a detailed overview on deploying and hardening OpenEMR in Cloud (AWS). The steps provided in this document will make your life easy in experiencing a hassle free and easy deployment of OpenEMR in AWS, in addition to enhancing and enforcing overall security of OpenEMR in Cloud thereby eliminating the risk of potential security threats.

2 Launching AWS Instance

2.1 User Account in AWS

To create AWS account, follow up with the below link:

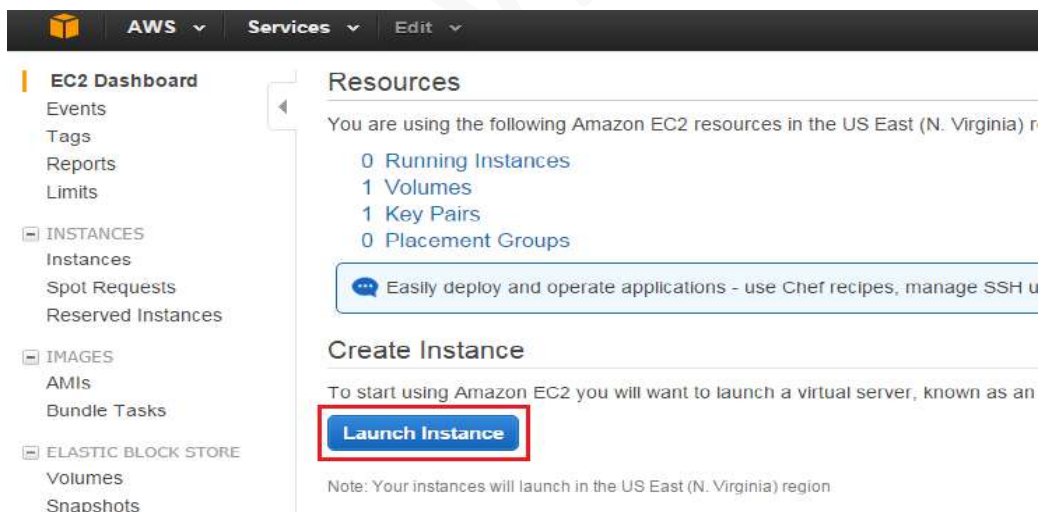
<https://aws.amazon.com/> and click "Create AWS Account".

2.2 Creating instance in AWS

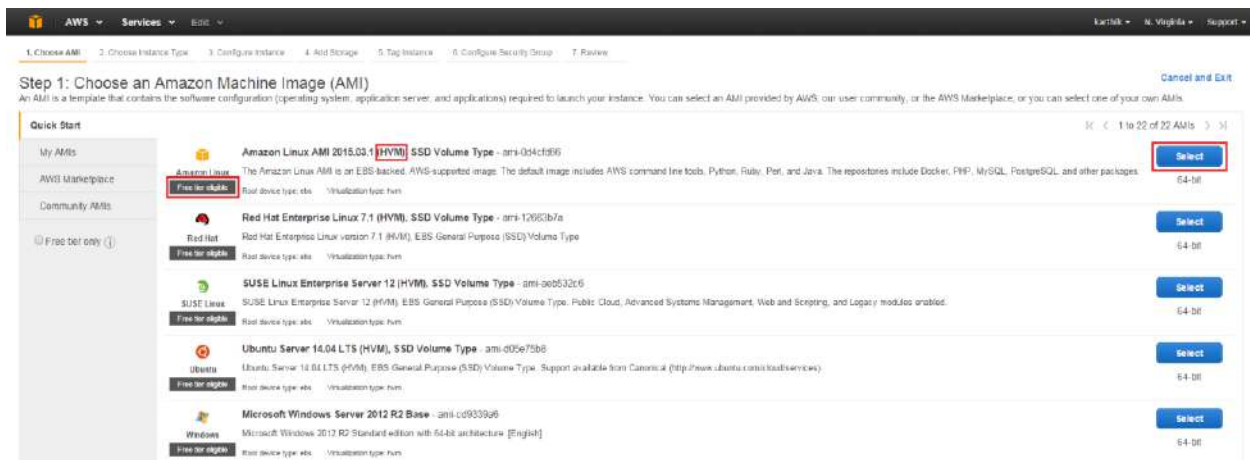
The following procedure is intended to help you to launch your first instance quickly and doesn't go through all possible options.

To launch a new instance,

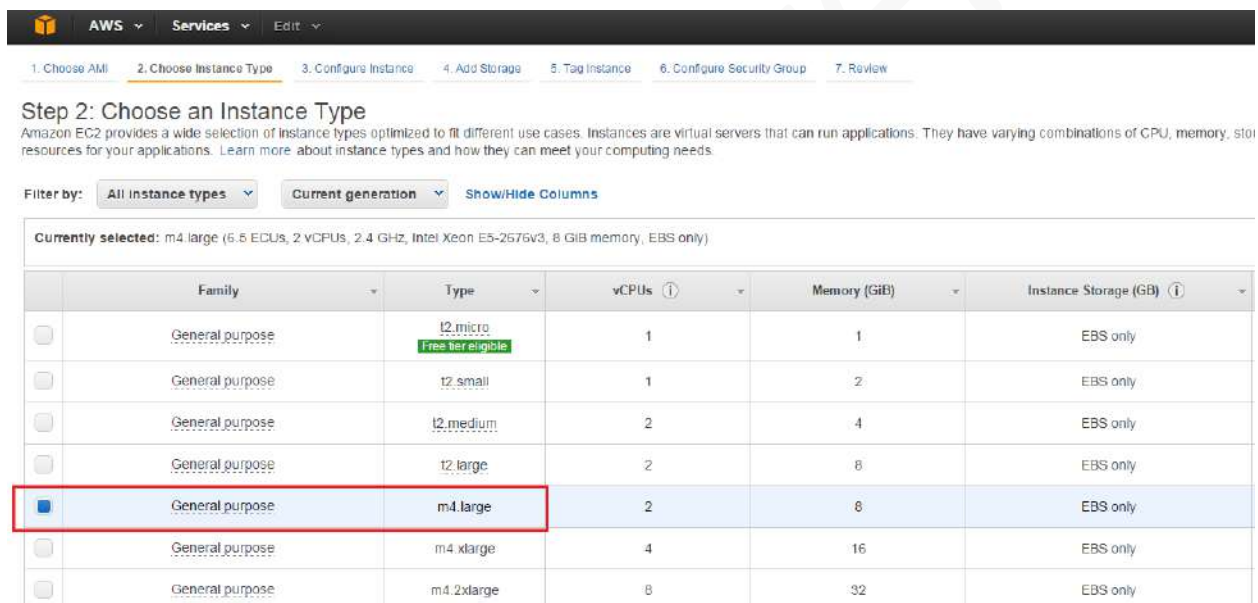
- Open the Amazon EC2 console at <https://console.aws.amazon.com/ec2/>
- Login using your AWS credentials
- Click EC2 under Compute category at console home.
- From the console dashboard, click **Launch Instance**.



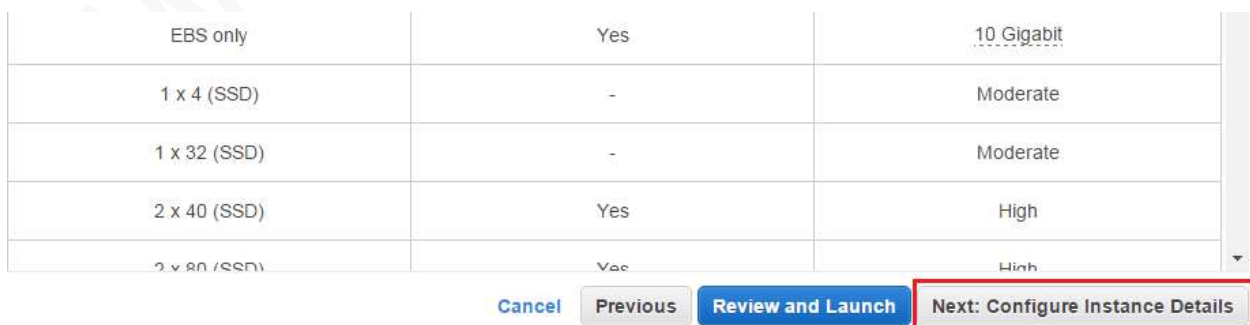
- **Choose an Amazon Machine Image (AMI)** page that displays a list of basic configurations called Amazon Machine Images (AMIs) that serve as templates for your instance. Select the HVM edition of the Amazon Linux AMI and click on the 'Select' button. Notice that this configuration is marked "Free tier eligible."



- On the **Choose an Instance Type** page, you can select the hardware configuration of your instance. Select one of the following instance types, which are the only instance types eligible for the free tier.
 - m4.large



- Click on **Next: Configure Instance Details**



- Make sure default options are selected for Network, Subnet and Auto-assign Public ip fields until unless you have new VPC

Step 3: Configure Instance Details

Configure the instance to suit your requirements. You can launch multiple instances from the same AMI, request Spot instances to take advantage of the lower pricing, assign an access management role to the instance, and more.

Number of instances: 1

Purchasing option: ☐ Request Spot Instances

Network: vpc-fd4ab099 (172.31.0.0/16) (default) [Create new VPC](#)

Subnet: No preference (default: subnet in any Availability Zone) [Create new subnet](#)

Auto-assign Public IP: Use subnet setting (Enable)

IAM role: None [Create new IAM role](#)

Shutdown behavior: Stop

[Cancel](#) [Previous](#) [Review and Launch](#) [Next: Add Storage](#)

- Next, click **Review and Launch** to let the wizard complete the other configuration settings for you.
- On the **Review Instance Launch** page, under **Security Groups**, you'll see that the wizard has created and selected a security group for you. Instead, select the security group that you had created during installation, using the below steps:
 - Click **Edit security groups**.
 - On the **Configure Security Group** page, ensure the **Select an existing security group** option is selected.
 - Select your security group from the list of existing security groups, and click **Review and Launch**.

Step 7: Review Instance Launch

Please review your instance launch details. You can go back to edit changes for each section. Click **Launch** to assign a key pair to your instance and complete the launch process.

Improve your instances' security. Your security group, launch-wizard-4, is open to the world.
Your instances may be accessible from any IP address. We recommend that you update your security group rules to allow access from known IP addresses only. You can also open additional ports in your security group to facilitate access to the application or service you're running, e.g., HTTP (80) for web servers. [Edit security groups](#)

AMI Details [Edit AMI](#)
Amazon Linux AMI 2016.03.1 (HVM), SSD Volume Type - ami-0c4cfd56
The Amazon Linux AMI is an EBS-backed, AWS-supported image. The default image includes AWS command line tools, Python, Ruby, Perl, and Java. The repositories include Docker, PHP, MySQL, PostgreSQL, and other packages.
Next Center Type: ami-0c4cfd56 Instance type: t1.micro

Instance Type [Edit instance type](#)

Instance Type	ECUs	vCPUs	Memory (GiB)	Instance Storage (GiB)	EBS-Optimized Available	Network Performance
t2.micro	Variable	1	1	EBS only	-	Low to Moderate

Security Groups [Edit security groups](#)

Instance Details [Edit instance details](#)

Storage [Edit storage](#)

Tags [Edit tags](#)

Step 6: Configure Security Group

A security group is a set of firewall rules that control the traffic for your instance. On this page, you can add rules to allow specific traffic to reach your instance. For example, if you want to set up a web server and allow internet traffic to reach your instance, add rules that allow unrestricted access to the HTTP and HTTPS ports. You can create a new security group or select from an existing one below. Learn more about Amazon EC2 security groups.

Assign a security group: ☐ Create a new security group

☒ Select an existing security group

Security Group ID	Name	Description	Actions
sg-0b17553c	default	default VPC security group	Copy to new
sg-263761d1	launch-wizard-1	launch-wizard-1 created 2015-05-04T19:51:17.595+05:30	Copy to new
sg-55952e32	launch-wizard-2	launch-wizard-2 created 2015-05-05T17:57:42.842+05:30	Copy to new
sg-e97c379e	launch-wizard-3	launch-wizard-3 created 2016-05-06T19:03:12.671+05:30	Copy to new

Inbound rules for sg-8b17553c (Selected security groups: sg-8b17553c)

Type	Protocol	Port Range	Source
All traffic	All	All	sg-0b17553c (default)

[Cancel](#) [Previous](#) [Review and Launch](#)

- On the **Review Instance Launch** page, click **Launch**.
- In the **Select an existing key pair or create a new key pair** dialog box, select **create a new key pair**. Select **Create a new key pair**, enter a name for the key pair, and then click **Download Key Pair**. This is the only chance for you to save the private key file, so be sure to download it. Save the private key file in a safe place. You'll need to provide the name of your key pair when you launch an instance and the corresponding private key each time you connect to the instance. A key pair enables you to connect to a Linux instance through SSH. Therefore, don't select the **Proceed without a key pair** option. If you launch your instance without a key pair, then you can't connect to it. When you are ready, select the acknowledgment check box, and then click **Launch Instances**.

Select an existing key pair or create a new key pair

A key pair consists of a **public key** that AWS stores, and a **private key file** that you store. Together, they allow you to connect to your instance securely. For Windows AMIs, the private key file is required to obtain the password used to log into your instance. For Linux AMIs, the private key file allows you to securely SSH into your instance.

Note: The selected key pair will be added to the set of keys authorized for this instance. Learn more about [removing existing key pairs from a public AMI](#).

Create a new key pair

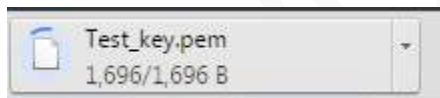
Key pair name
Test_key

Download Key Pair

You have to download the **private key file** (*.pem file) before you can continue. **Store it in a secure and accessible location.** You will not be able to download the file again after it's created.

Cancel Launch Instances

- (.pem) key will be downloaded.



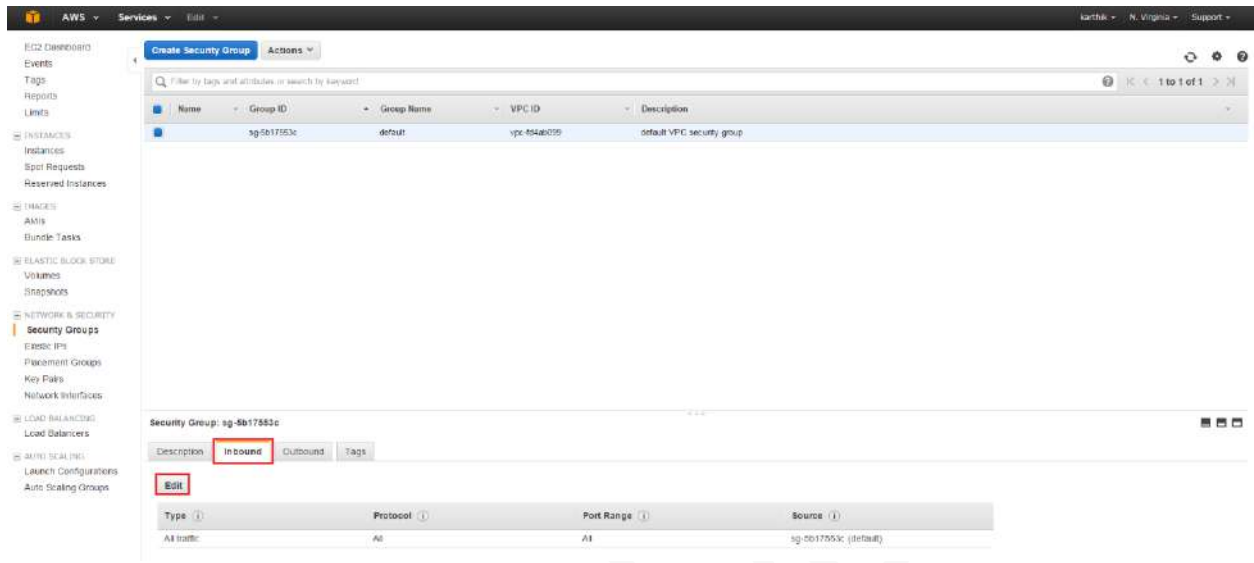
- A confirmation page lets you know that your instance is launching. Click **View Instances** to close the confirmation page and return to the console.
- On the **Instances** screen, you can view the status of your instance. It takes a short time for an instance to launch. When you launch an instance, its initial state is pending. After the instance starts, its state changes to running, and it receives a public DNS name. (If the **Public DNS** column is hidden, click the **Show/Hide** icon and select **Public DNS**.)

Name	Instance ID	Instance Type	Availability Zone	Instance State	Status Checks	Alarm Status	Public DNS	Public IP	Key Name	Monitoring	Launch Time	Security
mydemo	i-9021e843	t2.micro	us-east-1a	running	Initializing	None	ec2-52-2-87-16.compute-1.amazonaws.com	52.2.87.15	Test_key	disabled	September 10, 2015 at 6:45	default
	i-9021e843	t2.micro	us-east-1a	terminated		None			karthik	disabled	September 7, 2015 at 7:37	

2.3 Create Security Groups

- Go to the **Security Group** settings in the left navigation

- Find the Security Group that your instance is a part of
- Select **Inbound** tab and click on **Edit**.



- Select SSH from Type drop down and specify the Source (port 22)
- Repeat the same step for HTTP (80) and HTTPS (443) by clicking **Add Rule**
- Click **Save**

Edit inbound rules

Type	Protocol	Port Range	Source
SSH	TCP	22	Anywhere 0.0.0.0/0
HTTP	TCP	80	Anywhere 0.0.0.0/0
HTTPS	TCP	443	Anywhere 0.0.0.0/0

Add Rule
Cancel
Save

2.4 Connecting the Instance from Windows Using PuTTY

2.4.1 Prerequisites

Before you connect to your Linux instance using PuTTY, complete the following prerequisites:

Install PuTTY - Download and install PuTTY in your windows machine. Be sure to install the entire suite.

Get the ID of the instance - You can get the ID of your instance using the Amazon EC2 console (from the Instance ID column).

Get the public DNS name of the instance - You can get the public DNS for your instance using the Amazon EC2 console (check the Public DNS column; if this column is hidden, click the Show/Hide icon and select Public DNS).

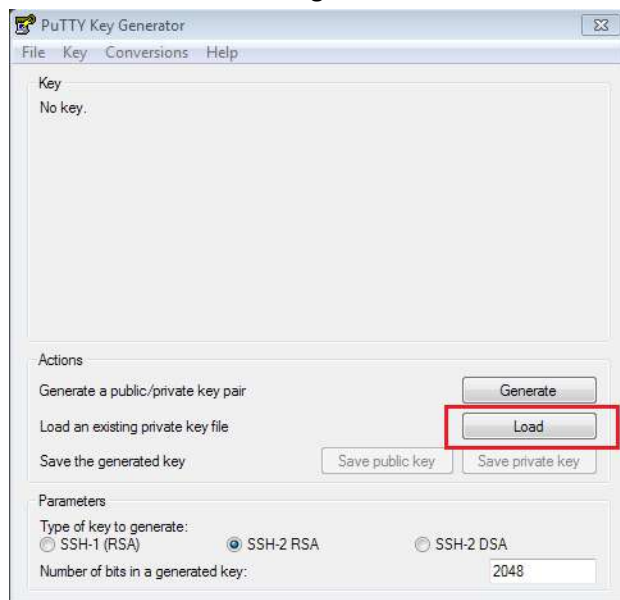
Name	Instance ID	Instance Type	Availability Zone	Instance State	Status Checks	Alarm Status	Public DNS	Public IP	Key Name	Monitoring	Launch Time	Security
	i-29fua3aa	t2.micro	us-east-1b	running	2/2 checks	None	ec-2-63-3-213-181.com...	52.3.213.181	Test_key	disabled	September 10, 2015 at 5:26	default
my-demos	i-9607e4d3	t2.micro	us-east-1a	terminated		None			karthik	disabled	September 7, 2015 at 7:37:4	

2.4.2 Convert Key PEM to PPK on Windows (Ignore for Linux Systems)

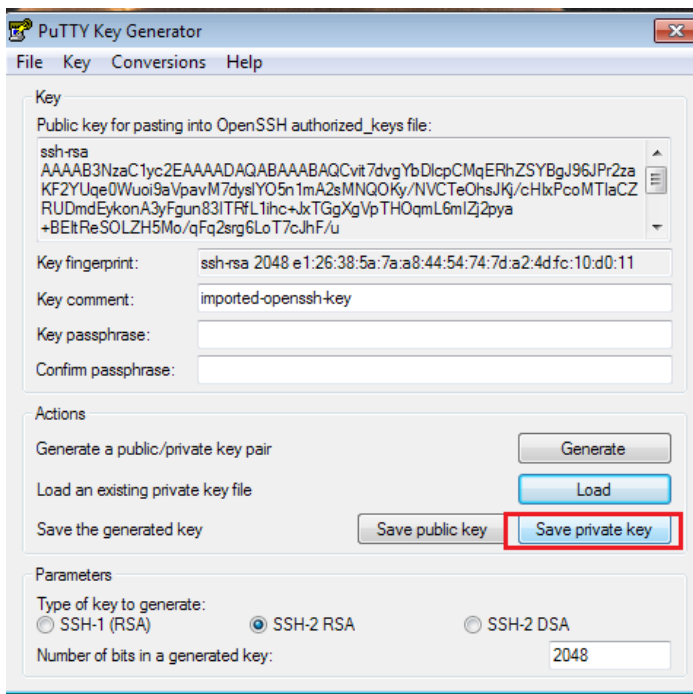
- Download your .pem from AWS (That has been already downloaded).



- Open **PuTTYgen**
- Click "**Load**" on the right side in the screen



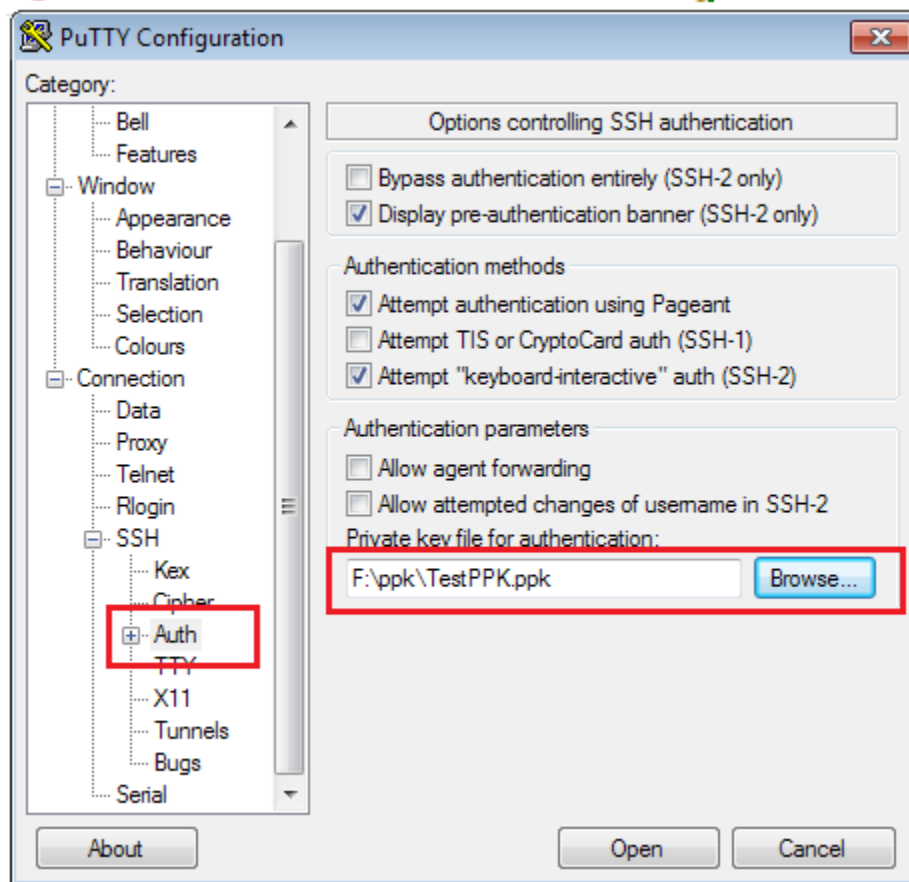
- Set the file type to All Files (*.*)
- Browse to, and Open your .pem file
- PuTTY will auto-detect everything it needs, and you just need to click "**Save private key**" and you can save your ppk key for use with PuTTY.



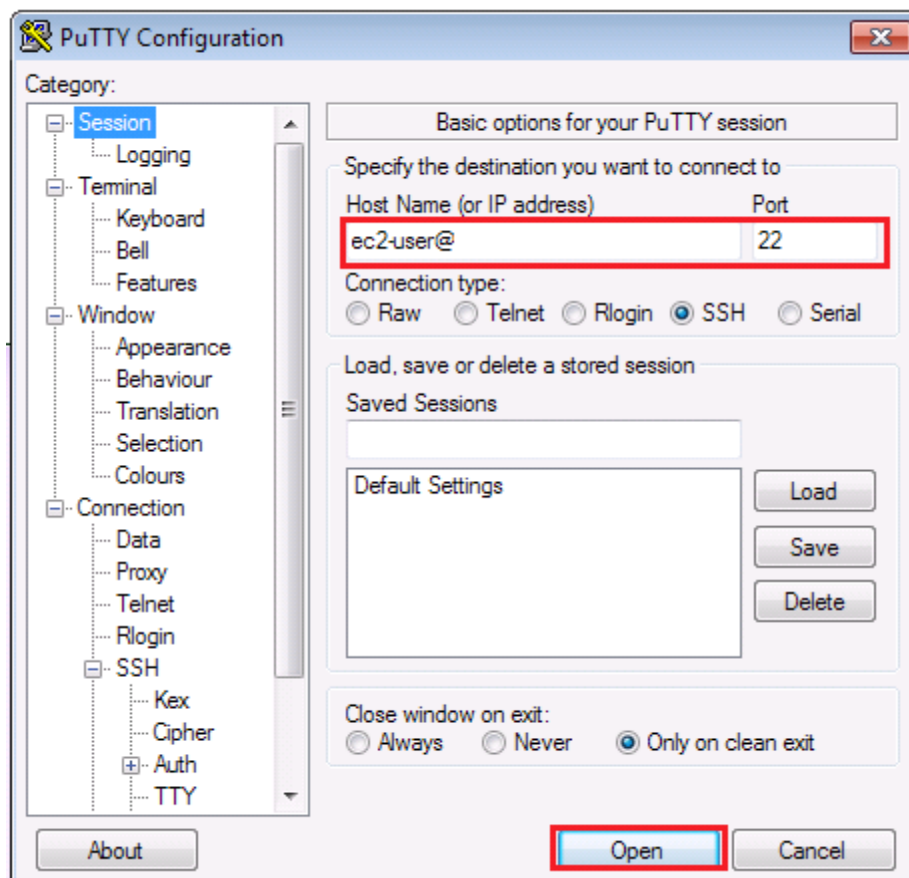
Note: Select Yes for saving without pass phrase (if alert box found).

2.4.3 Connect your AWS Instance using PUTTY :

- Start PuTTY (from the Start menu by clicking All Programs -> PuTTY -> PuTTY).
- In the Category pane, expand Connection, expand SSH, and then select Auth. Complete the following:
 - Click Browse.
 - Select the .ppk file that you generated for your key pair, and then click Open.
 - (Optional) If you plan to start this session again later, you can save the session information for future use. Select Session in the Category tree, enter a name for the session in Saved Sessions, and then click Save.
 - Click Open to start the PuTTY session.



- In the Category pane, select **Session** and complete the following fields:
 - In the Host Name box, enter user_name@public_dns_name. Be sure to specify the appropriate user name for your AMI. For example:
 - For an Amazon Linux AMI, the user name is ec2-user.
 - For a RHEL5 AMI, the user name is either root or ec2-user.
 - For an Ubuntu AMI, the user name is ubuntu.
 - For a Fedora AMI, the user name is either fedora or ec2-user.
 - For SUSE Linux, the user name is either root or ec2-user.
 - Otherwise, if ec2-user and root don't work, check with the AMI provider.
 - Under Connection type, select SSH.
 - Ensure that Port is 22.



- If this is the first time you have connected to this instance, PuTTY displays a security alert dialog box that asks whether you trust the host you are connecting to.
- (Optional) Verify that the fingerprint in the security alert matches the fingerprint that you obtained in step 1. If these fingerprints don't match, someone might be attempting a "man-in-the-middle" attack. If they match, continue to the next step.
- Click Yes. A window opens and you are connected to your instance.

3 Installing MySQL, Apache and PHP

Once you login to the ssh terminal, execute the following command to install apache, Mysql and PHP

```
# yum install httpd mysql mysql-server php php-mysql
```

Start httpd and mysql service,

```
# service httpd start
```

```
# service mysql start
```

Install git for code versioning and to checkout latest changes to the installed instances.

```
#yum install git
```

4 Installing OpenEMR

Go to web root directory, most probably it could be either one of following two

```
/var/www/
```

```
/var/www/html
```

Execute the following command to download OpenEMR community code from github.

```
# git clone https://github.com/openemr/openemr.git openemr
```

The above code will create a directory with source code in the name of openemr. Now, in browser provide url as <http://localhost/openemr>

Following screen will appear

OpenEMR Setup

Welcome to OpenEMR. This utility will step you through the installation and configuration of OpenEMR for your practice.

- Before proceeding, be sure that you have a properly installed and configured MySQL server available, and a PHP configured webserver.
- Detailed installation instructions can be found in the ['INSTALL'](#) manual file.
- If you are upgrading from a previous version, do NOT use this script. Please read the 'Upgrading' section found in the ['INSTALL'](#) manual file.

We will now ensure correct file and directory permissions before starting installation:

Ensuring following files are world-writable...

UNABLE to open file '/var/www/dhofmemr/sites/default/sqlconf.php' for writing.
(configure file permissions; see below for further instructions)

UNABLE to open file '/var/www/dhofmemr/interface/modules/ZendModules/config/application.config.php' for writing.
(configure file permissions; see below for further instructions)

You can't proceed until all above files are ready (world-writable).

In linux, recommend changing file permissions with the 'chmod 666 filename' command.

Fix above file permissions and then click the 'Check Again' button to re-check files.

Check Again

Provide necessary permission (in this case it is 666) to the mentioned file and click on 'Check Again' button. When all the necessary files get the appropriate permission, you will see the below screen.

OpenEMR Setup

Welcome to OpenEMR. This utility will step you through the installation and configuration of OpenEMR for your practice.

- Before proceeding, be sure that you have a properly installed and configured MySQL server available, and a PHP configured webserver.
- Detailed installation instructions can be found in the ['INSTALL'](#) manual file.
- If you are upgrading from a previous version, do NOT use this script. Please read the 'Upgrading' section found in the ['INSTALL'](#) manual file.

We will now ensure correct file and directory permissions before starting installation:

Ensuring following files are world-writable...

'/var/www/dhofmemr/sites/default/sqlconf.php' file is **ready**.

'/var/www/dhofmemr/interface/modules/ZendModules/config/application.config.php' file is **ready**.

Ensuring following directories have proper permissions...

'/var/www/dhofmemr/sites/default/documents' directory is **ready**.

'/var/www/dhofmemr/sites/default/edi' directory is **ready**.

'/var/www/dhofmemr/sites/default/era' directory is **ready**.

'/var/www/dhofmemr/sites/default/letter_templates' directory is **ready**.

'/var/www/dhofmemr/gacl/admin/templates_c' directory is **ready**.

'/var/www/dhofmemr/interface/main/calendar/modules/PostCalendar/pntemplates/compiled' directory is **ready**.

'/var/www/dhofmemr/interface/main/calendar/modules/PostCalendar/pntemplates/cache' directory is **ready**.

All required files and directories have been verified. Click to continue installation.

Continue

Click continue to next step to set up database. Choose the option 'Have setup create the database' for fresh installation setup and click continue.

OpenEMR Setup

Step 1

Now I need to know whether you want me to create the database on my own or if you have already created the database for me to use. For me to create the database, you will need to supply the MySQL root password.

NOTE: clicking on "Continue" may delete or cause damage to data on your system. Before you continue please backup your data.

- ☒ Have setup create the database
☐ I have already created the database

Continue

Enter the database name and credentials for the same in the following screen. Also, enter the username and password for the default admin user. Make sure to note the same. Click on continue to proceed to next screen.

OpenEMR Setup

Step 2

Now you need to supply the MySQL server information and path information. Detailed instructions on each item can be found in the [INSTALL](#) manual file.

MYSQL SERVER:

Server Host:	localhost	(If you run MySQL and Apache/PHP on the same computer, then leave this as 'localhost'. If they are on separate computers, then enter the IP address of the computer running MySQL.)
Server Port:	3306	(This is the MySQL port. The default port for MySQL is 3306.)
Database Name:	openemrdemo	(This is the name of the OpenEMR database in MySQL - 'openemr' is the recommended)
Login Name:	openemrdemo	(This is the name of the OpenEMR login name in MySQL - 'openemr' is the recommended)
Password:		(This is the Login Password for when PHP accesses MySQL - it should be at least 8 characters long and composed of both numbers and letters)
Name for Root Account:	root	(This is name for MySQL root account. For localhost, it is usually ok to leave it 'root'.)
Root Pass:		(This is your MySQL root password. For localhost, it is usually ok to leave it blank.)
User Hostname:	localhost	(If you run Apache/PHP and MySQL on the same computer, then leave this as 'localhost'. If they are on separate computers, then enter the IP address of the computer running Apache/PHP)
UTF-8 Collation:	General	(This is the collation setting for mysql. Leave as 'General' if you are not sure. If the language you are planning to use in OpenEMR is in the menu, then you can select it. Otherwise, just select 'General'.)

OPENEMR USER:

Initial User:	admin	(This is the login name of user that will be created for you. Limit this to one word.)
Initial User Password:		(This is the password for the initial user account above.)
Initial User's First Name:	Administrator	(This is the First name of the 'initial user'.)
Initial User's Last Name:	Administrator	(This is the Last name of the 'initial user'.)
Initial Group:	Default	(This is the group that will be created for your users. This should be the name of your practice.)

Click on continue if you get all 'OK' for creation of database and other stuff as shown below.

OpenEMR Setup

Step 3

Configuring OpenEMR...

Connecting to MySQL Server... OK.
Creating database... OK.
Creating user with permissions for database... OK.
Reconnecting as new user... OK.
Creating Main tables... OK
Creating Language Translation (utf8) tables... OK
Creating CVX Immunization Codes tables... OK
Writing SQL configuration... OK.
Setting version indicators... OK
Writing global configuration defaults... OK
Adding Initial User... OK

Next step will install and configure access controls (php-GACL).

As step 4, Access controls will be configured. Click on continue to next step.

OpenEMR Setup

Step 4

Installing and Configuring Access Controls (php-GACL)...

Configuration:

driver = **mysql**,
host = **localhost:3306**,
user = **openemrdemo**,
database = **openemrdemo**,
table prefix = **gac**l_Testing database connection...
Success! Connected to "**mysql**" database on "**localhost:3306**".
Testing database type...
Success! Compatible database type "**mysql**" detected!
Making sure database "**openemrdemo**" exists...
Success! Good, database "**openemrdemo**" already exists!
Success! First Step of Access Control Installation Successful!!!

OpenEMR ACL Setup

All done configuring and installing access controls (php-GACL)! Gave the 'admin' user (password is 'pass') administrator access.

Done installing and configuring access controls (php-GACL).

Next step will configure PHP.

Continue

Make a note on path of php.ini file existence for future references and click on continue to proceed further.

OpenEMR Setup

Step 5

Configuration of PHP...

We recommend making the following changes to your PHP installation, which can normally be done by editing the php.ini configuration file:

- Your **php.ini** file can be found at **/etc/php5/apache2/php.ini**
- To ensure proper functioning of OpenEMR you must make sure that settings in php.ini file include "short_open_tag = On", "display_errors = Off", "register_globals = Off", "max_execution_time" set to at least 60, "max_input_time" set to at least 90, "post_max_size" set to at least 30M, and "memory_limit" set to at least "128M".
- In order to take full advantage of the patient documents capability you must make sure that settings in php.ini file include "file_uploads = On", that "upload_max_filesize" is appropriate for your use and that "upload_tmp_dir" is set to a correct value that will work on your system.

We recommend you print these instructions for future reference.

Next step will configure Apache web server.

Continue

Append the information provided in step 6 to apache configuration file. Apache configuration file will be found at

/etc/httpd/conf/httpd.conf

Once you are done with appending the information, click on continue.

OpenEMR Setup

Step 6

Configuration of Apache web server...

The `/var/www/dhofmemr/sites/*/documents`, `/var/www/dhofmemr/sites/*/edi` and `/var/www/dhofmemr/sites/*/era` directories contain patient information, and it is important to secure these directories. Additionally, some settings are required for the Zend Framework to work in OpenEMR. This can be done by pasting the below to end of your apache configuration file:

```
<Directory "/var/www/dhofmemr">
    AllowOverride FileInfo
</Directory>
<Directory "/var/www/dhofmemr/sites">
    AllowOverride None
</Directory>
<Directory "/var/www/dhofmemr/sites/*/documents">
    order deny,allow
    Deny from all
</Directory>
<Directory "/var/www/dhofmemr/sites/*/edi">
    order deny,allow
    Deny from all
</Directory>
<Directory "/var/www/dhofmemr/sites/*/era">
    order deny,allow
    Deny from all
</Directory>
```

If you are having difficulty finding your apache configuration file, then refer to the ["INSTALL"](#) manual for suggestions.

We recommend you print these instructions for future reference.

Click 'continue' for further instructions.

Continue

You are at the last step. Make a note of user name and password for default admin user if you missed at step 2.

OpenEMR Setup

Congratulations! OpenEMR is now installed.

- Access controls (php-GACL) are installed for fine-grained security, and can be administered in OpenEMR's admin->acl menu.
- Reviewing `/var/www/dhofmemr/library/classes/.../sites/default/config.php` is a good idea. This file contains some settings that you may want to change.
- There's much information and many extra tools bundled within the OpenEMR installation directory. Please refer to `openemr/Documentation`. Many forms and other useful scripts can be found at `openemr/contrib`.
- To ensure a consistent look and feel through out the application using [Firefox](#) is recommended.
- The OpenEMR project home page, documentation, and forums can be found at <http://www.open-emr.org>
- We pursue grants to help fund the future development of OpenEMR. To apply for these grants, we need to estimate how many times this program is installed and how many practices are evaluating or using this software. It would be awesome if you would email us at president@oemr.org if you have installed this software. The more details about your plans with this software, the better, but even just sending us an email stating you just installed it is very helpful.

We recommend you print these instructions for future reference.

The initial OpenEMR user is 'admin' and the password is 'pass'

If you edited the PHP or Apache configuration files during this installation process, then we recommend you restart your Apache server before following below OpenEMR link.

[Click here to start using OpenEMR.](#)

Click on the link "Click here to start using OpenEMR" and you will screen the login page of OpenEMR. Enter the credentials of default admin user and get logged in successfully!



5 System security check after OpenEmr Installation:

5.1.1 Securing System Ports

Log into terminal and check opened ports in the system,
Sudo netstat -tulp

If any unwanted ports is opened, then stop those respective services.

Sudo chkconfig --list |grep on

To permanently disable Network Ports and Disabling Runlevel System Services **e.g** the runlevel service nfs, run:
Sudo chkconfig nfs off

To immediately disable the runlevel service nfs, run:
Sudo /etc/init.d/nfs stop

5.1.2 Securing SSH

Default configuration for SSH meets the security requirements for many environments. However, there are a few parameters in **/etc/ssh/sshd_config** that you want to change,

It's prudent to disable direct root logins at the SSH level
PermitRootLogin no

Preventing Accidental Denial of Service.
UsePrivilegeSeparation yes

Since SSH protocol version 1 is not as secure you may want to limit the protocol to version 2 only

Protocol 2

You may also want to prevent SSH from setting up TCP port and X11 forwarding if you don't need it:

AllowTcpForwarding no

X11Forwarding no

Ensure the StrictModes directive is enabled which checks file permissions and ownerships of some important files in the user's home directory like ~/.ssh, ~/.ssh/authorized_keys etc. If any checks fail, the user won't be able to login

StrictModes yes

Ensure that all host-based authentications are disabled. These methods should be avoided as primary authentication

IgnoreRhosts yes

HostbasedAuthentication no

RhostsRSAAuthentication no

Disable sftp if it's not needed:

#Subsystem sftp /usr/lib/misc/sftp-server

After changing any directives make sure to restart the sshd daemon:

/etc/init.d/sshd restart

5.1.3 Kernel Tunable Security Parameters

Kernel parameters you can use to secure your Linux server against attacks.

Config file **/etc/sysctl.conf** to make the change permanent after reboots. To activate the configured kernel parameters immediately at runtime, use:

sysctl -p

Enable TCP SYN Cookie Protection

net.ipv4.tcp_syncookies = 1

Disable IP Source Routing

net.ipv4.conf.all.accept_source_route = 0

Disable ICMP Redirect Acceptance

net.ipv4.conf.all.accept_redirects = 0

Enable IP Spoofing Protection

net.ipv4.conf.all.rp_filter = 1

Enable Ignoring to ICMP Requests

```
net.ipv4.icmp_echo_ignore_all = 1
```

Enable Ignoring Broadcasts Request

```
net.ipv4.icmp_echo_ignore_broadcasts = 1
```

Check Kernel Tunable Security Parameters by issuing the below commands,
`Sysctl -p`

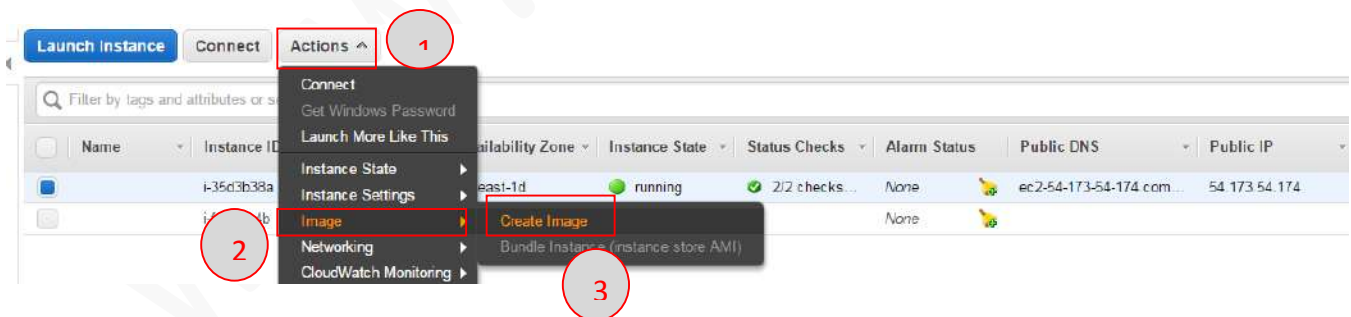
Check opened ports, if any un-necessary ports are opened then follow the step (7)
`Netstat -tlup`

To ensure available ports use nmap from your local machine,
`nmap -sTU <OpenEmr_config_machine>`

6 Golden Image:

Create golden image for the instance with the pre-configured OpenEMR settings,

To create golden Image, select the instance go to Actions, Image and create image,



In Create Image dialog box, provide the image name & give the description for reference,

Create Image

Instance ID ⓘ

i-35d3b38a

Image name ⓘ

ViSolve-OpenEMR ⓘ

Image description ⓘ

Golden Image for OpenEMR Testing ⓘ

No reboot ⓘ

☐

Instance Volumes

Type ⓘ	Device ⓘ	Snapshot ⓘ	Size (GiB) ⓘ	Volume Type ⓘ	IOPS ⓘ	Delete on Termination ⓘ	Encrypted ⓘ
Root	/dev/xvda	snap-9be40de6	8	General Purpose (SSD) ▾	24 / 3000	☒	Not Encrypted

Add New Volume

Total size of EBS Volumes: 8 GiB
When you create an EBS image, an EBS snapshot will also be created for each of the above volumes.

Cancel

Create Image ⓘ

Create Image

✓ Create Image request received.

[View pending image ami-47134022](#)

Any snapshots backing your new EBS image can be managed on the [snapshots screen](#) after successful image creation.

Close

Image will be created and you can use this image for new instance creation with preconfigured settings.

6.1.1 Launch OpenEMR Instance with Golden Image

To Launch AWS instance with the created Golden Image, Go to Launch instance (as mentioned in section-3 '[Launching AWS instance](#)') select My AMIs tab under 'step 1.Choose AMI' and use the created image as like below,

AWS ServicesEdit

karthikN. VirginiaSupport

1. Choose AMI2. Choose Instance Type3. Configure Instance4. Add Storage5. Tag Instance6. Configure Security Group7. Review

Step 1: Choose an Amazon Machine Image (AMI)

[Cancel and Exit](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

Quick Start

My AMIs

AWS Marketplace

Community AMIs



ViSolve-OpenEMR - ami-47134022
Golden Image for OpenEMR Testing
Root device type: ebs Virtualization type: hvm Owner: 808340445281

Select

64-bit

And follow the same procedure to complete the setup.

Hardening OpenEMR in Cloud (AWS)

Last Updated on November 2015

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